Solution Requirements

Date	17 September 2026
Team ID	SWTID-2026-3976
Project Name	AI FAQ Assistant API (AnswerBot)
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

The requirements below define what the AI FAQ Assistant API does and how well it must perform.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users register with name, email and password (min. 6 characters, unique email). Passwords are hashed with bcrypt. Login returns a signed JWT used as a Bearer token on protected routes.	High
2	Authorization levels	Public users can view and search FAQs only. Authenticated users can create FAQs, use the AI endpoints and update or delete the FAQs they authored (createdBy check). Admin and editor roles are defined for later phases.	High
3	External interfaces	Google Gemini API through the @google/genai SDK for AI answers and FAQ generation; MongoDB through Mongoose; API clients such as Postman or ThunderClient.	High
4	Transactions processing	FAQ create, read, update and delete operations persisted in MongoDB with Mongoose schema validation. AI output is generated first and only stored when the author saves it as an FAQ.	High
5	Reporting	Every endpoint returns a consistent JSON response (success, message, data). Search returns matching FAQ lists. Usage analytics and dashboards are planned for a later phase.	Medium
6	Business rules, etc.	A FAQ must have a question, answer and category; category must be one of Technology, Education, Health, Banking or General. Only the original author may modify or delete an FAQ. AI routes require login to prevent abuse.	High
7	Compliance to laws or regulations	User passwords are never stored in plain text or returned in responses; secrets are kept in a .env file. No personal data beyond name and email is collected.	Medium
8	Other	Centralised error handling that maps duplicate keys, validation errors and invalid IDs to clean HTTP status codes without leaking database details.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Standard API calls should respond quickly; AI calls depend on Gemini latency.	Auth and FAQ requests under 2 s; AI answers under 5 s (observed: 48-664 ms and 2.56 s)
2	Scalability	Stateless JWT sessions and layered modules (routes, controllers, services) allow horizontal scaling; MongoDB Atlas can be scaled independently.	Runs behind a load balancer without session stickiness

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
3	Security & Data Privacy	bcrypt hashing, JWT route protection, input validation and sanitisation, CORS and secrets in environment variables.	No plain-text passwords; protected routes return 401 without a valid token
4	Reliability & Availability	Global error handler and unhandled-rejection handling keep the server stable.	Server never crashes on bad input; consistent error JSON
5	Usability & Accessibility	Clear REST endpoints, descriptive validation messages and a Postman collection for easy testing.	All endpoints documented and testable from Postman
6	Other	Maintainability through MVC-style separation and portability (Node.js 18+, any OS).	Runs on Windows, macOS and Linux with npm install